ANSWER KEY

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LEARN · PLAY · WONDER

FACT CHECK · C6 · ELECTRICITY

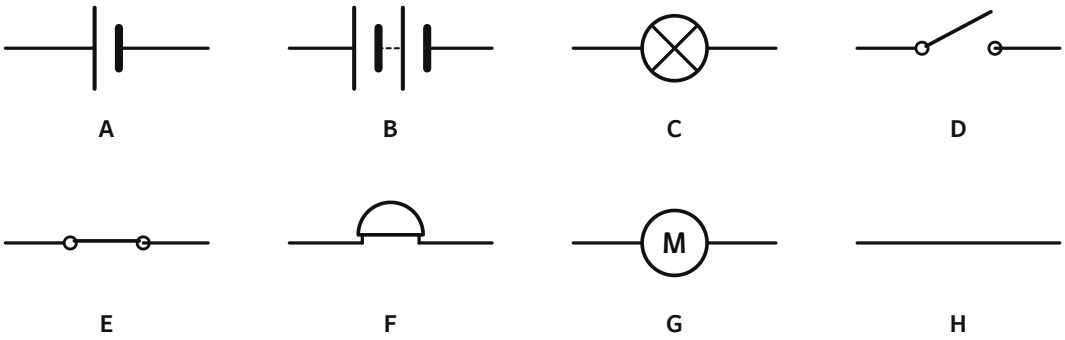
Circuits and symbols

★☆☆
Grade 4

Learning goals

- Use circuit symbols.
- Sort conductors and insulators.

1 Name each symbol. [8]



A	B	C	D	E	F	G	H
cell	battery	lamp	open switch	closed switch	buzzer	motor	wire

2 Sort the materials. [8]

SORT THESE

wooden stick **pencil lead (graphite)** **steel spoon** **copper** **glass**
aluminium foil **rubber band** **plastic ruler**

Conductor	Insulator
copper	plastic ruler
steel spoon	rubber band
aluminium foil	wooden stick
pencil lead (graphite)	glass

3 **Will the bulb light? Write yes or no and explain.** [4]

Circuit	Lights?	Why?
cell, bulb and wires in a complete loop	yes	the circuit is complete
one wire is not connected	no	there is a gap
a plastic straw joins the gap	no	plastic is an insulator
a metal key joins the gap	yes	metal is a conductor

ANSWER KEY

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THINK IT THROUGH · C6 · ELECTRICITY

Predict and explain

★★☆
Grade 4

Learning goals

- Predict changes to circuits.
- Explain safety features.

1 **Draw a circuit diagram with two cells, one lamp and a switch. Use the correct symbols.** [3]

My circuit

A rectangle loop with two cell symbols (or a battery), a lamp (circle with a cross) and a switch.

2 **Predict and explain.** [6]

Change	Prediction	Reason
add a second bulb	both bulbs dimmer	the same cell shares its push between two bulbs
add a second cell	bulb brighter	more push makes more electricity flow
make the wire much longer	a little dimmer	a long wire lets less electricity through

3 **Why is the inside of a cable made of copper and the outside made of plastic?** [2]

Copper conducts electricity well; plastic is an insulator that keeps the electricity in and protects us.

4 **Why is it dangerous to use a hair dryer with wet hands in the bathroom?** [1]

Water with dirt or salt can conduct electricity, so electricity could flow through your body and give a dangerous shock.

ANSWER KEY

INVESTIGATE · C6 · ELECTRICITY

Does wire length change brightness?

★★☆ Grade 4

Learning goals

- Plan a fair test.
- Measure with a light meter.
- Draw and use a line graph.

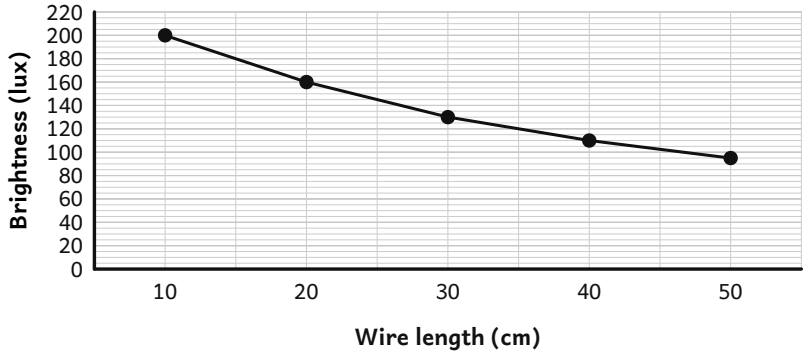
Class 4C made a circuit with a cell, a bulb and a long piece of thin nichrome wire. They moved a crocodile clip along the wire to change the length in the circuit, and used a light meter app to measure the brightness of the bulb in lux.

1 **Complete the plan.** [3]

Change	length of the wire in the circuit
Measure	brightness of the bulb (lux)
Keep the same	the same cell, bulb and type of wire; the meter at the same distance; a dark room

2 **Draw a line graph.** [3]

Wire length (cm)	10	20	30	40	50
Brightness (lux)	200	160	130	110	95



3 **Describe the pattern.** [2]

The longer the wire, the dimmer the bulb.

4 **Predict the brightness for 60 cm.** [1]

about 80–85 lux

5 **A dimmer switch for a lamp works in a similar way. How do you think it works?** [1]

It changes how much wire (resistance) the electricity has to go through.

ANSWER KEY

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REAL-WORLD CASE STUDY · C6 · ELECTRICITY

Lights for every home

★★★
Grade 4

Learning goals

- Use data about electricity in Bangladesh.
- Think about electrical safety.
- Plan how to save electricity.

Science | History | Maths | Citizenship

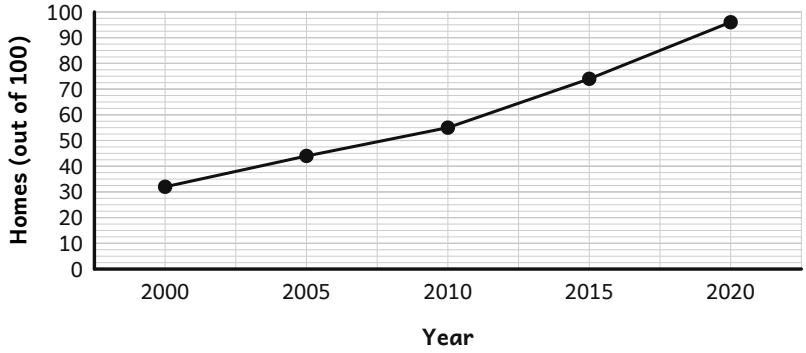
In the year 2000, only about **one in three** homes in Bangladesh had electricity. Children studied by the light of kerosene lamps, which were smoky and dangerous. Since then, new power stations, power lines and millions of **solar home systems** have brought electricity to almost **every home** in the country.

Electricity has changed lives: children can study at night, farmers use electric pumps, shops stay open and people charge mobile phones. But more electricity also means more need for **safety** and for **saving energy**, because much of it is made by burning gas and oil.

Approximate figures from World Bank data on access to electricity in Bangladesh.

1 **Homes with electricity in Bangladesh (approximate).** [3]

Year	2000	2005	2010	2015	2020
Homes with electricity (out of 100)	32	44	55	74	96



a) Draw the line graph.

b) By how much did it rise from 2000 to 2020?

96 – 32 = 64 more homes out of every 100

2 **Give two ways electricity has helped families.** [2]

Children can study at night with bright, safe light; fans and pumps; charging phones; shops and clinics can work.

3 **Spot the danger. What is wrong in each case and how should it be fixed?** [3]

Situation	Danger and fix
six plugs in one socket with an adaptor	overloading can overheat and start a fire; use fewer plugs
a cable with the plastic cover cracked	bare wire can give a shock; replace the cable
a kite caught on a power line	never pull it down; electricity can travel down the string; tell an adult

4 **Write three tips for saving electricity at home.** [2]

switch off lights and fans when leaving; use LED bulbs; unplug chargers; open windows for light and air.

ANSWER KEY

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REVIEW AND CHALLENGE · C6 · ELECTRICITY

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

A	C	W	N	K	B	J	I	W	P
V	J	O	S	I	U	R	F	I	L
L	E	T	P	G	Z	R	S	R	A
C	F	C	F	P	Z	C	W	E	S
I	I	B	E	E	E	R	I	C	T
M	E	R	U	L	R	R	T	C	I
O	G	F	C	L	L	K	C	V	C
T	D	P	D	U	B	O	H	O	Y
O	E	L	O	M	I	H	A	U	S
R	U	K	O	D	A	T	C	I	I

bulb buzzer cell circuit copper motor plastic switch wire

2 Which material is a conductor? [1]

- ○ rubber
- ○ plastic
- ● **copper**
- ○ wood

3 What happens if you add a second bulb to a series circuit? [1]

- ○ both bulbs are brighter
- ● **both bulbs are dimmer**
- ○ nothing changes
- ○ the cell explodes

4 Design a burglar alarm for a box: when the lid opens, a buzzer sounds. Draw the circuit. [3]

Box alarm

Cell, buzzer and a switch made of foil strips; opening the lid closes the switch and completes the circuit.

5 **Challenge. Why is it important to have a switch in a circuit for a motor that turns a fan?** [2]

So we can turn the fan on and off safely without touching wires; we can stop it quickly if fingers or clothes get caught.